

Dhia Problem 3 — Revised Results (Response to Comments 1–4)

Comment 1: Task 1 table redone with $\sigma_2 = 0$.

Comment 2: dual solutions obtained for $\lambda > 0$ (Task 1, $\lambda = 1$).

Comment 3: parameter changed (Gr) so the effect of X on $f''(0)$ and $-t'(0)$ is clearly visible.

Comment 4: profiles start at $\eta = 0$ on the x-axis, combine $X = 0, \pi/4, \pi/2$ in every profile, and combine the nanofluid and dust phases for velocity and temperature respectively.

Comments 1 & 2 — Task 1 with $\sigma_2 = 0$, both solutions

$\lambda = 1, \phi_1 = \phi_2 = \beta_v = \beta_T = s = X = 0, Pr = 6.2, Ec = 1, m = 1, \varepsilon = 1, \sigma_2 = 0$ and $Gr = 0$ (the program's residual $Gr = 0.001$ shifted $f''(0)$ in the 4th decimal; with $Gr = 0$ the no-slip case gives $f''(0) = -1.000000$ exactly and every σ_1 matches Hayat et al., Ibrahim & Shankar and Anuar et al. — see comparison below). A second solution exists for this stretching case ($\lambda = 1 > 0$) across the whole σ_1 range shown: it was located with the program's short-domain initial guess and traced with σ_1 as the free continuation parameter (branch fold slightly below $\sigma_1 = 0$). It is a velocity-undershoot state — f' dips below zero in mid-layer and the entrainment $f(\infty)$ is much smaller (0.129 vs 0.839 at $\sigma_1 = 0.5$).

	first solution				second solution			
sigma1	f''(0)	F'(0)	-t'(0)	tp(0)	f''(0)	F'(0)	-t'(0)	tp(0)
0.0	-1.000000	0.000000	2.451133	0.000000	-1.022343	-0.000000	2.095833	0.000000
0.1	-0.872082	0.000000	2.544436	0.000000	-0.889867	0.000000	3.212430	0.000000
0.2	-0.776377	0.000000	2.587516	0.000000	-0.791127	-0.000000	2.671361	0.000000
0.5	-0.591195	0.000000	2.594800	0.000000	-0.597705	0.000000	2.551187	0.000000
2.0	-0.283980	0.000000	2.290811	0.000000	-0.287587	-0.000000	2.292908	0.000000
5.0	-0.144840	0.000000	1.904145	0.000000	-0.146522	0.000000	1.849608	0.000000

Benchmark comparison for $f''(0)$ (first solution)

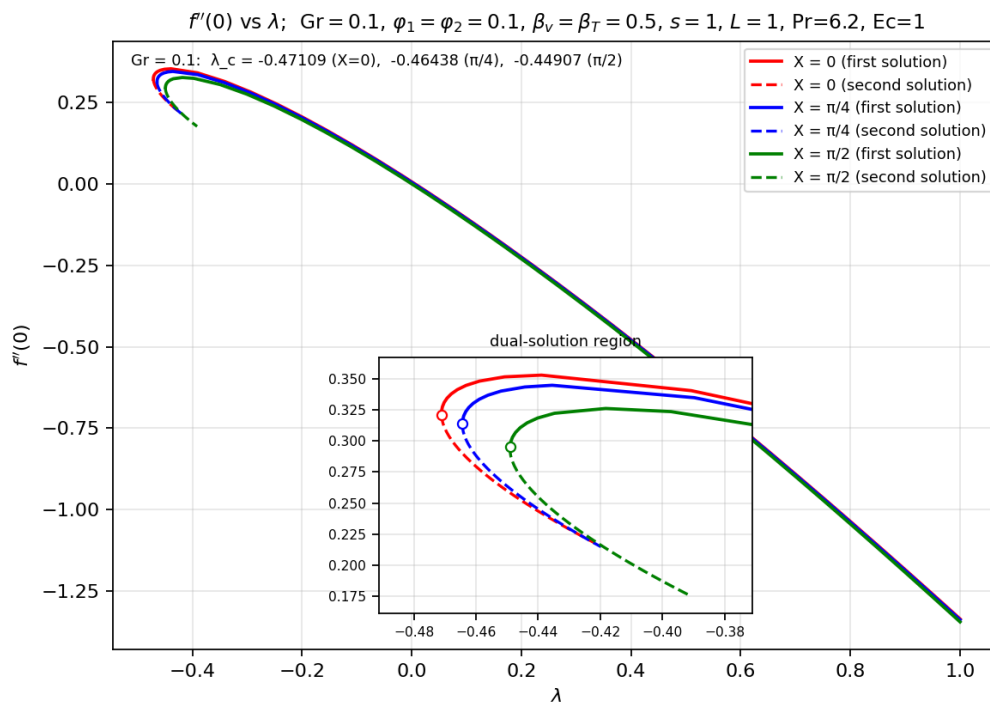
sigma1	Hayat et al. [36]	Ibrahim & Shankar [37]	Anuar et al. [33]	Present (Gr=0)
0.0	-1.000000	-1.0000	-1.000000	-1.000000
0.1	-0.872082	-0.8721	-0.872083	-0.872082
0.2	-0.776377	-0.7764	-0.776377	-0.776377
0.5	-0.591195	-0.5912	-0.591196	-0.591195
2.0	-0.283981	-0.2840	-0.283981	-0.283980
5.0	-0.144841	-0.1448	-0.144842	-0.144840

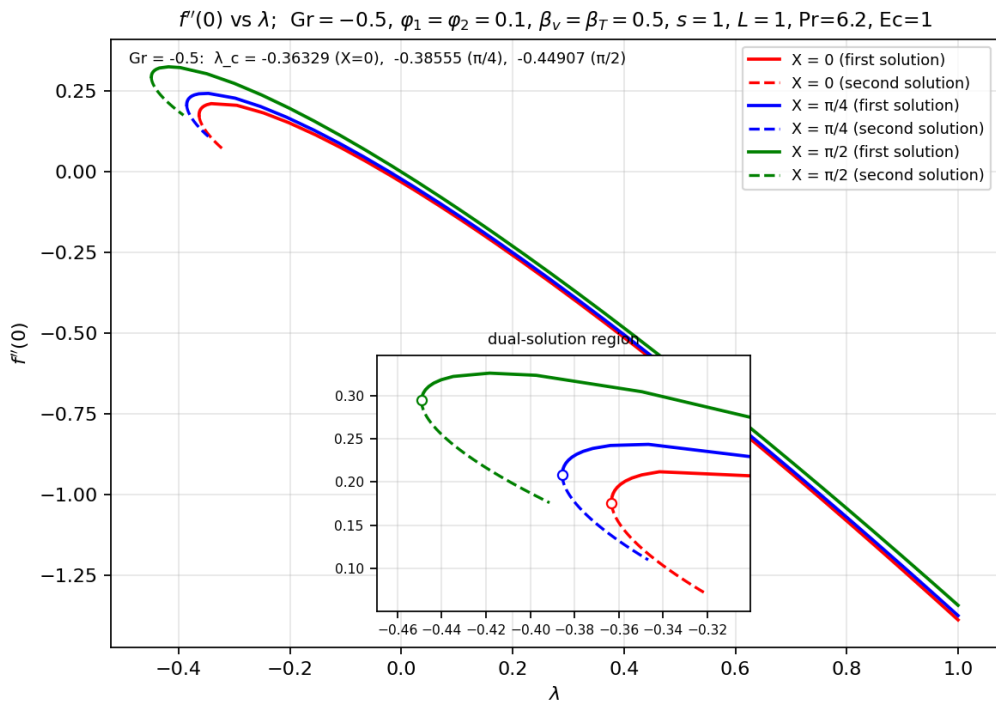
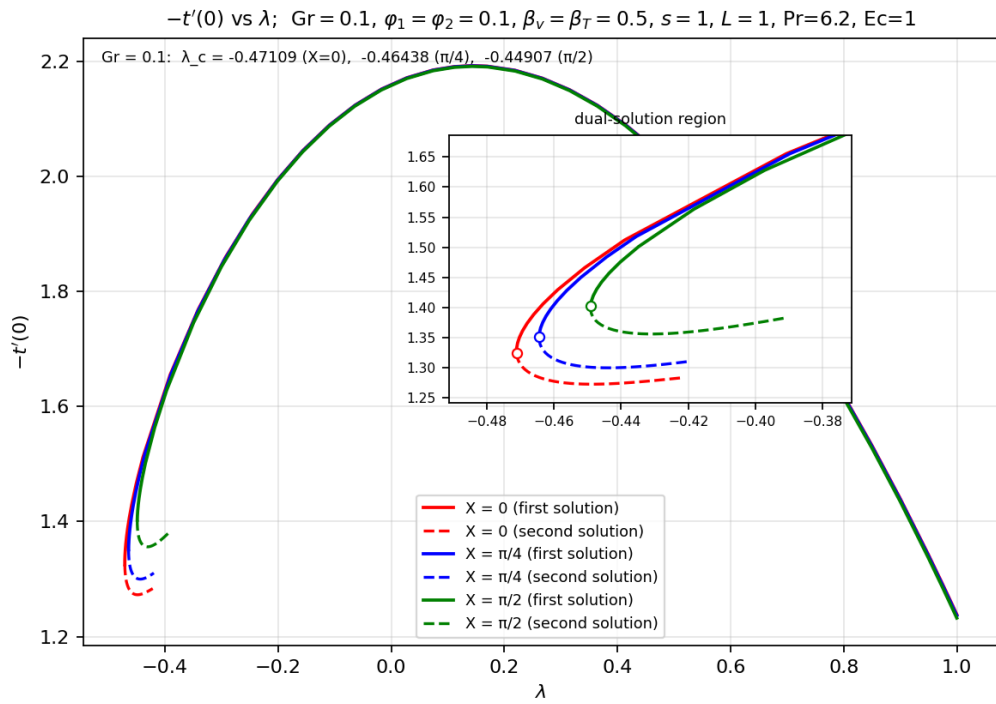
With $\beta_v = \beta_T = 0$ the dust phase decouples, so $F'(0) = t_p(0) = 0$ exactly on both branches. With $\sigma_2 = 0$ the wall temperature is fixed, $t(0) = 1$. **Important caveat on the second branch:** unlike the first solution (exponential far-field decay, values independent of $\eta_{\max}$), the second branch decays only *algebraically* ($f' \sim \eta^{-2}$), which is the classical non-uniqueness of the stretching-sheet problem. Its wall values therefore depend weakly on the truncation: second-solution values above are computed at $\eta_{\max} = 16$ (inside the 8–15 range the assignment prescribes for etaMax2) and drift slowly as the domain grows — e.g. at $\sigma_1 = 0, f''(0) = -1.0223$ ($\eta_{\max} = 16$) $\rightarrow$ -1.0124 ($\eta_{\max} = 50$). This is also why the MATLAB program's second solution is sensitive to the choice of etaMax2. (Short-domain runs additionally produce spurious large- $|t'(0)|$ “solutions” near thermal eigen-resonances; those were excluded by verification.)

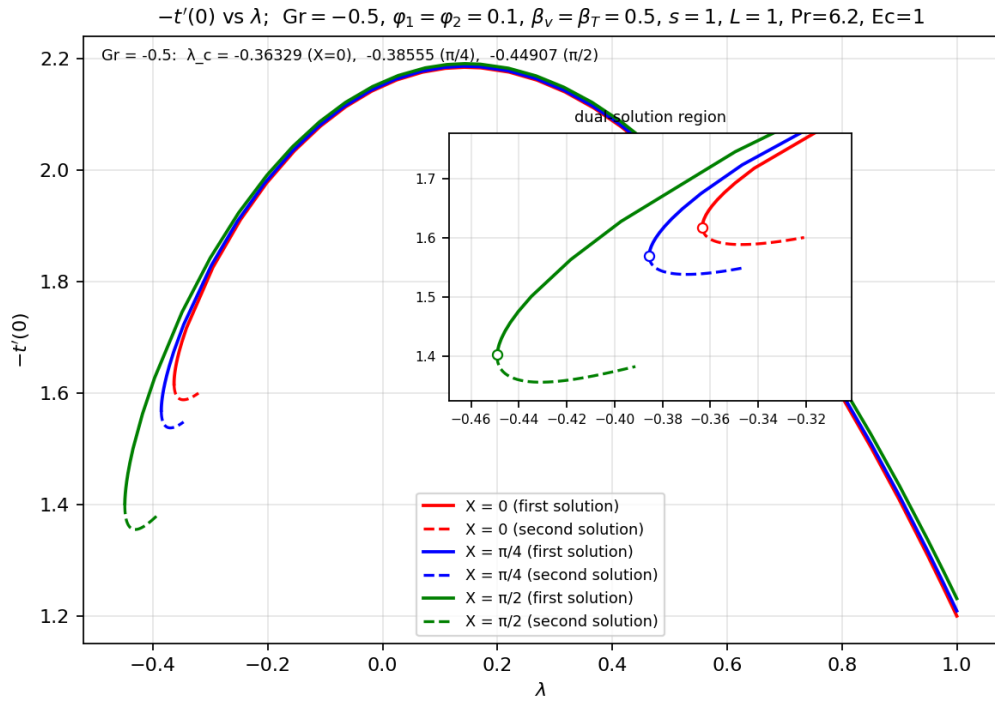
Comment 3 — Gr raised so the inclination X has a clear effect

X enters only through $Gr \cdot A_8 \cdot t \cdot \cos(X)$; the program's $Gr = 0.001$ made the three X curves coincide. **Gr is the parameter changed** (all task-prescribed parameters kept). For $Gr \geq 0.2$ the $Ec = 1$ dissipation term causes thermal runaway before any fold, so two revised sets are shown: $Gr = 0.1$ (largest assisting value with complete dual structures for all three X) and $Gr = -0.5$ (opposing flow, strongest separation). $X = \pi/2$ is identical for every Gr since $\cos(\pi/2) = 0$.

Set	Gr	lambda_c (X=0)	lambda_c (X=pi/4)	lambda_c (X=pi/2)	Spread
Baseline (program)	0.001	-0.44928	-0.44922	-0.44907	0.0002
Revised (primary)	0.1	-0.47109	-0.46438	-0.44907	0.0220
Revised (opposing)	-0.5	-0.36329	-0.38555	-0.44907	0.0858

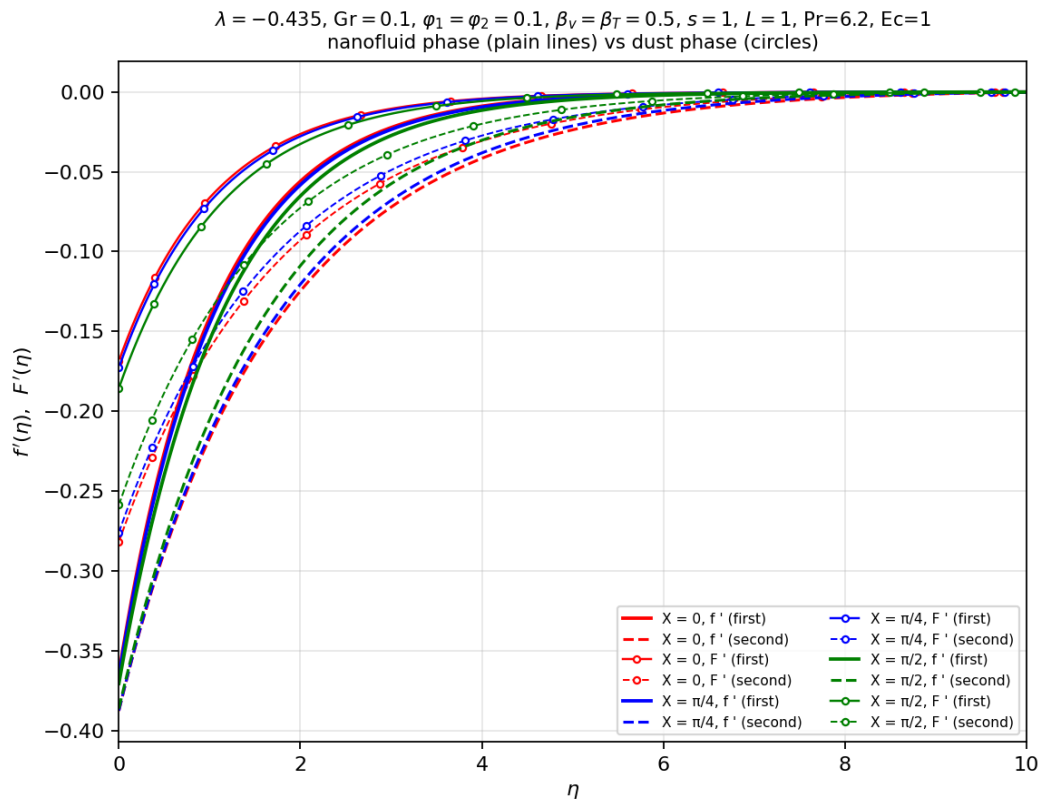




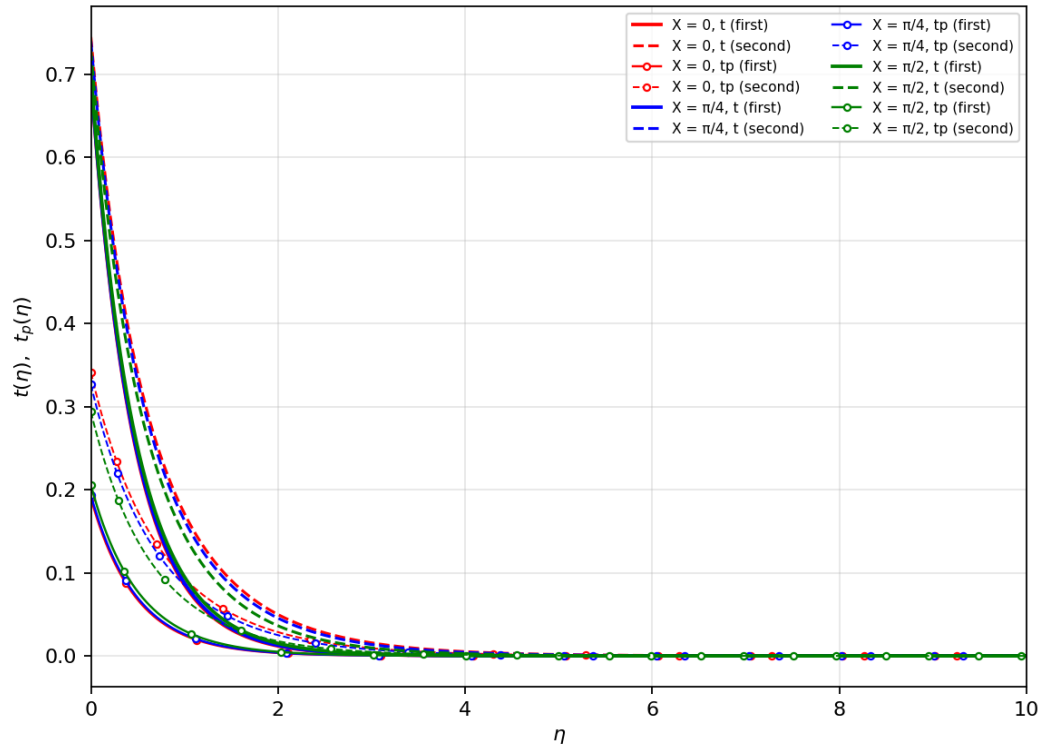


Comment 4 — combined profiles (x-axis from 0; $X = 0$, $\pi/4$, $\pi/2$; nanofluid + dust phases)

Dual-solution profiles at $Gr = 0.1$ and $\lambda = -0.435$, chosen inside the common dual-solution window of all three inclinations. Solid = first solution, dashed = second solution; red/blue/green = $X = 0$, $\pi/4$, $\pi/2$. In the phase-combined figures the plain lines are the nanofluid phase (f' , t) and the circle-marked lines the dust phase (F' , t_p). Every x-axis starts exactly at $\eta = 0$. All eight OdeBC boundary conditions are satisfied and all profiles decay to zero.

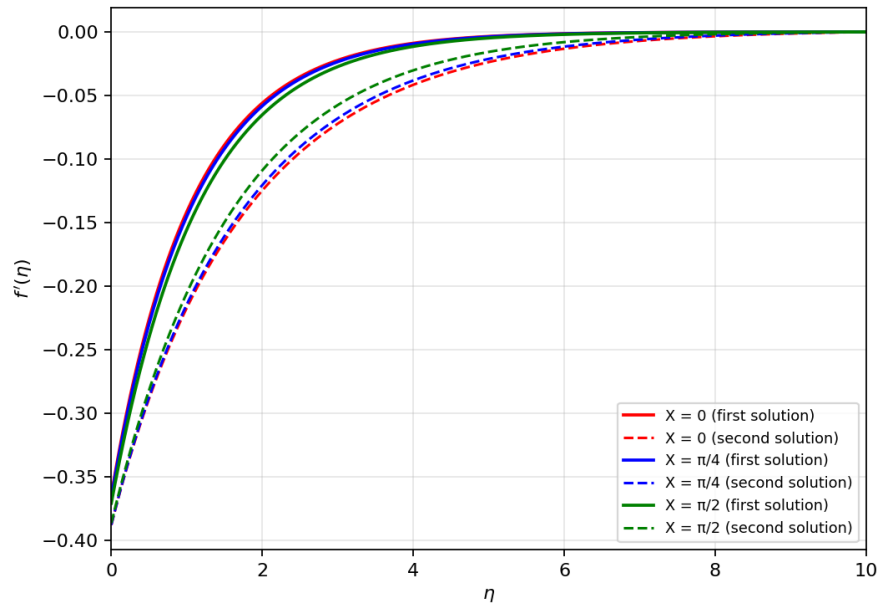


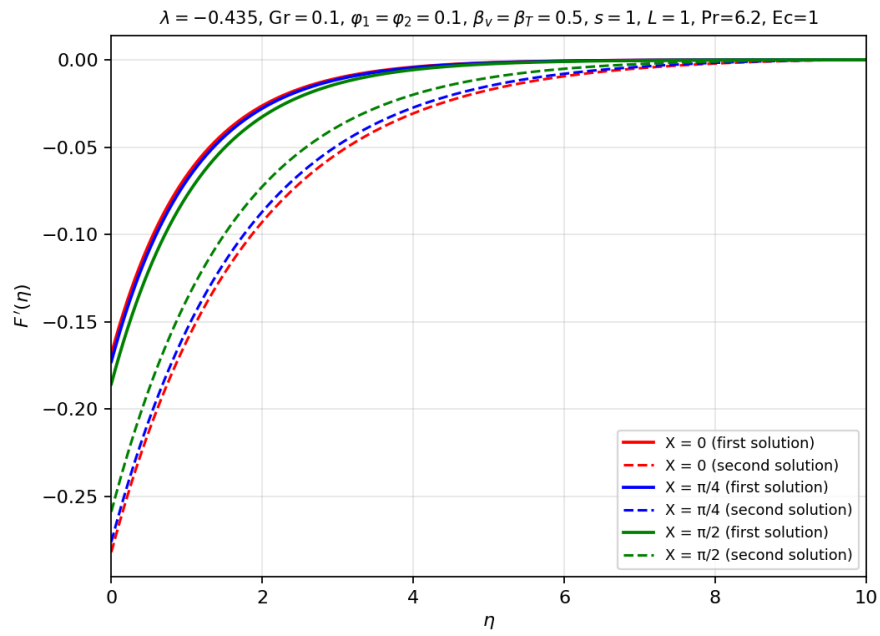
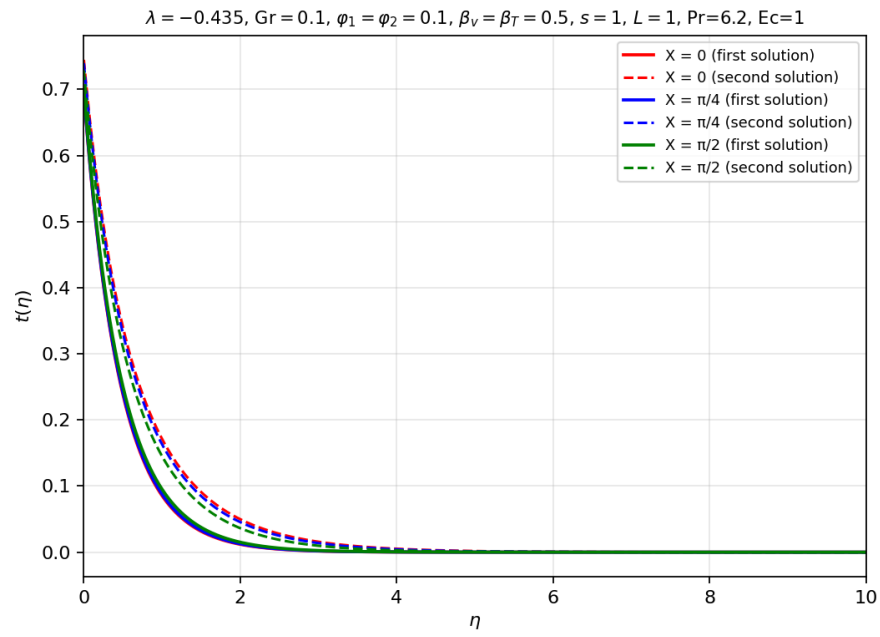
$\lambda = -0.435$, $Gr = 0.1$, $\phi_1 = \phi_2 = 0.1$, $\beta_v = \beta_T = 0.5$, $s = 1$, $L = 1$, $Pr = 6.2$, $Ec = 1$
nanofluid phase (plain lines) vs dust phase (circles)

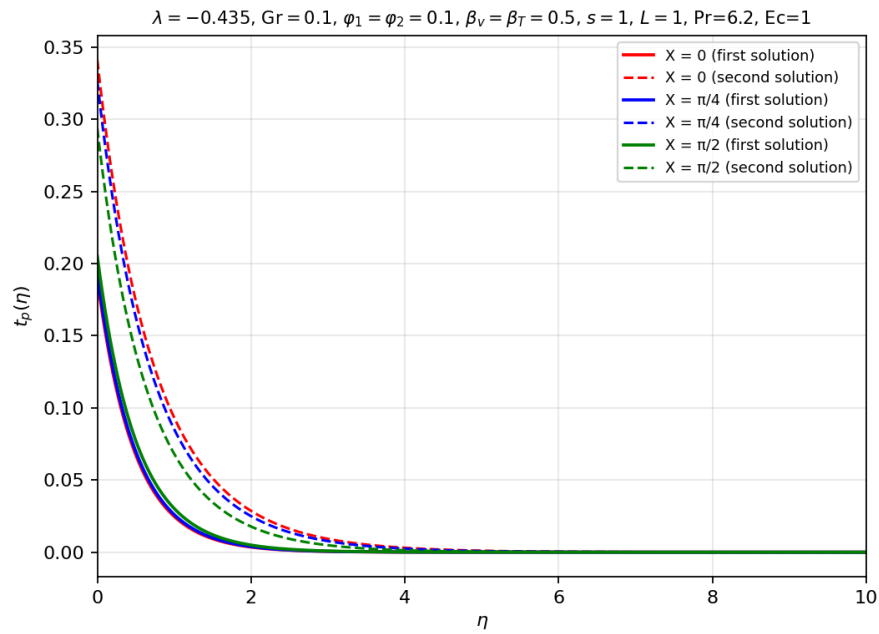


Per-quantity combined-X profiles

$\lambda = -0.435$, $Gr = 0.1$, $\phi_1 = \phi_2 = 0.1$, $\beta_v = \beta_T = 0.5$, $s = 1$, $L = 1$, $Pr = 6.2$, $Ec = 1$







Wall values at $\lambda = -0.435, Gr = 0.1$ (both solutions)

X	solution	$f''(0)$	$F'(0)$	$-t'(0)$	$tp(0)$
0	first	0.352639	-0.168565	1.525715	0.189810
0	second	0.236145	-0.281869	1.276313	0.341203
$\pi/4$	first	0.344665	-0.172713	1.519625	0.193491
$\pi/4$	second	0.237129	-0.275977	1.301748	0.326582
$\pi/2$	first	0.322030	-0.185560	1.500065	0.205360
$\pi/2$	second	0.243699	-0.258742	1.356687	0.293800

Full data in this folder: task1_table_sigma1_v2.csv, task2_curve_*.csv, task3_profile_*_X*.csv.